Single View Might Be Enough: A Task-Dependent Study of Multi-Perspective Necessity in Geosensing

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Abstract. Pairing overhead (satellite) and street-level imagery is widely assumed to help vision-language models reason about cities. We test this assumption directly. We build a multi-city visual question answering benchmark of 40 cities in which each location carries a satellite tile and four street-level views, and ask four-way multiple-choice questions about eight urban attributes derived from OpenStreetMap. Using a controlled image ablation—no image, satellite only, street only, and both—we measure *fusion synergy*, the accuracy a model gains from both views over its better single view. Across a 9B model we fine-tune on the data, four off-the-shelf remote-sensing VLMs, two city splits, and three prompting strategies including extended chain-of-thought, the gain is marginal: the two-view condition exceeds the better single view by only about two to three points on average, an order of magnitude below the 40–70 points the *same* models gain on cross-view questions that require both images by construction. A paired item-level analysis sharpens the picture: the two-view model trails a per-item single-view oracle by 8–9 points, and supplying the second view corrects a wrong answer on roughly 1% of items while flipping a correct answer to wrong on roughly 10%. For these urban-attribute tasks at commonly served resolutions, the second view contributes far less than its acquisition cost, and multi-perspective necessity is task-dependent rather than universal.

Keywords: Geosensing · Vision-language models · Multi-view fusion

1 Introduction

A recurring premise in geospatial machine learning is that more perspectives are better. Overhead imagery captures layout, footprints, and connectivity; ground imagery captures facades, surfaces, signage, and street furniture. Pairing the two is intuitively complementary, and a growing line of work collects, aligns, and jointly models satellite and street-level views for urban understanding. The implicit expectation is that a model with access to both views should outperform one restricted to either.

We ask whether this expectation actually holds for concrete urban-attribute recognition, and find that—for the tasks we study—it largely does not. We construct a multi-city visual question answering (VQA) benchmark in which every location is represented by a marked satellite tile and four cardinal street-level views, and pose four-way multiple-choice questions about eight urban attributes (land use, building height, urban density, road type, road surface, junction type, amenity richness, and transit density) grounded in OpenStreetMap (OSM) metadata [18]. Crucially, the same imagery also supports a family of *cross-view* questions that are unanswerable from a single view by construction (e.g. "does this street-level photo belong to this satellite tile?"). These cross-view tasks serve as a built-in positive control: any sound measurement of view complementarity must register a large benefit there.

Our central instrument is a controlled image ablation. For each question we evaluate a model under four input conditions—$\textsf{blind}$ (no image), $\textsf{sat}$ (satellite only), $\textsf{sv}$ (street only), and $\textsf{full}$ (both)—and define *fusion synergy* as $\textsf{full} - \max(\textsf{sat}, \textsf{sv})$: the accuracy gained from having both views over the better single view. Positive synergy is the signature of genuine cross-view fusion. We complement synergy with *vision contribution*, $\textsf{full} - \textsf{blind}$, which isolates how much the imagery contributes at all over a text-only prior.

The result is consistent across conditions and, we argue, robustly measured. On the eight attribute tasks the second view's contribution is marginal: across a 9B-parameter model we fine-tune on the benchmark, four off-the-shelf remote-sensing VLMs evaluated zero-shot, two distinct city splits (random per-city and held-out unseen cities), and three prompting strategies (plain decoding, a 16k-token explicit reasoning budget, and a four-turn observe–reason–self-check–commit protocol), having both views improves on the better single view by only about two to three points on average. A paired item-level analysis (Sec. 5.5) makes the size of this effect concrete: the two-view model trails a per-item single-view *oracle*—one that simply picks the better of the two single-view answers for each question—by 8–9 points, and the joint views contain essentially no information beyond their better single view. The mechanism is visible in the per-item flips: adding the second view rescues a wrong answer on about 1% of items but overturns a correct one on about 10%, so the model appears unable to adjudicate when the two views disagree. Vision itself clearly matters—$\textsf{full} - \textsf{blind}$ is roughly +10 points—so the model *is* reading the images; it simply derives a single view's worth of signal from them. And on the cross-view control tasks the very same models gain 40–70 points from the second view, demonstrating that the ablation detects fusion when fusion is genuinely required.

We frame this as a *task-dependent* observation, not a claim that overhead imagery is uninformative or that multi-view fusion is futile in general. It is a measured property of single-view-decidable attribute recognition at the satellite resolutions commonly served in practice. We make this scope explicit throughout, report bootstrap confidence intervals on every synergy estimate, audit the benchmark for text-only leakage, and characterize the items where the second view changes the answer.

Our contributions are:

1. A multi-city satellite + street VQA benchmark (40 cities, eight OSM-grounded urban-attribute tasks plus cross-view control tasks) with a clean location-level held-out split and a controlled four-way image ablation.
2. A measurement of multi-view fusion synergy that is near zero for all eight attribute tasks and robust across model scale, training regime, city split, and prompting strategy—with the cross-view tasks as a positive control that proves the measurement is sensitive.
3. An item-level analysis (per-item oracle ceilings, paired significance and equivalence tests, and a characterization of the view-flip set) that bounds the *latent* complementarity available to any fusion method, reframing "no method helped" as "there was little to capture."
4. Practical guidance: for these attribute families, serving a single (typically cheaper and better-covered) view costs essentially no accuracy, and a reusable ablation protocol for deciding when a second view is warranted.

2 Related Work

Remote-sensing vision-language models are single-view. A wave of instruction-tuned VLMs target remote sensing, including GeoChat [7], SkySenseGPT [15], LHRS-Bot and LHRS-Bot-Nova [9, 17], EarthGPT [28], and the multi-sensor generalist EarthDial [20]. These models are overhead-only: each consumes a single aerial or satellite image, and none ingests ground-level imagery (Tab. 2). Where "multi-modal" appears in this line of work it refers to multiple *overhead* sensors—optical, SAR, and infrared, as in EarthGPT—rather than to pairing satellite with street-level views. The dominant evaluation suites follow suit: RSVQA [14], VRSBench [8], and GeoChat's own benchmark all pose questions over a single overhead image. None isolates the marginal value of a second, ground-level view, which is the question we study. Architecturally these models build on LLaVA-style designs [11, 12] with CLIP or SigLIP encoders [19, 27]; we use four as zero-shot baselines and discuss their training-data provenance in Sec. 4.

Multi-view urban benchmarks. A smaller set of benchmarks does pair overhead and ground views for urban understanding. UrBench [30] evaluates multimodal LLMs on coordinate-aligned street and satellite pairs across fourteen task types, and reports that models perform *worse* on its cross-view tasks than on either single view—a pattern its authors attribute to a cross-view reasoning deficit, and which we revisit here for the narrower case of attribute recognition. UrbanLLaVA [3] trains a multi-modal model over satellite, street-view, OSM, and trajectory inputs; notably its attribute-style tasks (land use, density) are posed from satellite alone, while its multi-view supervision targets matching and localization. CityCube [25] benchmarks cross-view *spatial* reasoning (orientation, perspective taking) rather than attribute recognition. Closest to our finding, the concurrent CityLens [13] benchmarks LVLMs on socioeconomic indicators from satellite and street-view and reports that street-view alone performs comparably to using both views and well above satellite alone—a redundancy observation in the same domain, which our controlled ablation, item-level oracle analysis, and positive control turn into a quantified and mechanistic account. Our contribution relative to this group is not a first paired-view dataset, but a controlled measurement of *when* the second view is needed, with a built-in sensitivity check.

Cross-view geo-localization: where both views are necessary. Pairing satellite and ground imagery is a well-established setting in cross-view geo-localization, where the goal is to match a ground photo to an overhead reference [2, 26, 31]. There the two views are jointly *necessary* by definition—one cannot localize the ground image without the overhead one—and methods succeed precisely by modeling the distinct-but-linkable geometric layout the two views share [29]. This is the regime our cross-view control tasks occupy, and the contrast with it is exactly our point: complementarity that is constitutive of localization is largely absent from single-view-decidable attribute recognition.

The marginal value of a second modality. That adding a modality need not help is known beyond geosensing. Wang *et al.* [22] show that under naive joint training the best single-modal network can outperform the multi-modal one, and Liang *et al.* [10] formalize multimodal interactions into redundancy, uniqueness, and synergy, demonstrating that low-synergy, strong-unimodal regimes are real and measurable; our "fusion synergy" is the behavioral analogue of their synergy term. Against this, theory shows multimodal learning can provably lower achievable risk [5], and several aerial+ground fusion systems report gains for recognition—a unified near/remote model for land use and building function [24], multi-view scene-classification datasets [16], and, most directly counter to our setting, Suel *et al.* [21], who find that fusing satellite and street-level imagery measurably improves regression of income and environmental deprivation. We engage this last result directly in Sec. 7: it differs from ours in task type (continuous regression versus four-way recognition), label source, and model (a trained CNN versus zero-shot and fine-tuned VLMs), and our finding refines rather than contradicts the broad "fusion helps" literature by identifying a task regime in which it does not.

3 The Benchmark

Construction. The benchmark pairs overhead and street-level imagery with multiple-choice questions about urban attributes, generated from OpenStreetMap (OSM) metadata [18]. Each *location* is sampled within one of 40 cities across 32 countries on six continents and is associated with (i) one satellite tile, (ii) four street-level views at fixed bearings (forward, backward, left, right), and (iii) OSM-derived attribute labels. Satellite imagery is sourced per-location by geography—sub-meter aerial orthophotos: ESRI World Imagery ($\sim$0.3 m/px to

0.5 m/px in cities) as the global default, NAIP (∼0.6 m/px) in the US, and IGN (∼0.2 m/px) in France, with a coarse Sentinel-2 fallback (∼10 m/px) reserved for locations that no high-resolution source covers. In the benchmark every overhead tile resolves to a sub-meter source (4775 ESRI, 305 NAIP, 160 IGN; the Sentinel-2 fallback was never triggered), so the redundancy we report below is measured against genuinely high-resolution overhead imagery rather than a coarse proxy. The satellite tile carries a marker indicating the queried location; the marker is part of the input condition and is held fixed across all ablation modes (we examine its effect in Sec. 6).

Tasks. We study eight *attribute* tasks—land_use, building_height, urban_density, road_type, road_surface, junction_type, amenity_richness, and transit_density—each posed as a four-way multiple-choice question with a single correct letter. These are the questions for which "do we need both views?" is well-posed: each is, in principle, answerable from imagery of a single location. The dataset additionally contains *cross-view* tasks (view–view matching at two difficulty levels, in binary and multiple-choice form, and a camera-direction task) whose answer lives in the *relationship* between the two views; these are unanswerable from a single view by construction and we use them only as a positive control (Sec. 5.2).

Splits. All splits operate at the location level so that no location's questions span two splits. We hold out 10% of locations *per city* as a benchmark test set; this set is identical across training regimes and is disjoint from training data (verified: zero question-id overlap). For training we use two strategies that share this benchmark: a *per-city* split (a further fraction of locations per city to validation) and a *seen/unseen* split (eight cities held out entirely to validation, to probe cross-city generalization). Question generation uses a fixed seed for option shuffling, and we keep a single question per (location, task) pair to avoid location-level inflation. Locations with fewer than four street-level views are discarded.

Leakage audit. Because the questions are templated, we audit how much of each task is solvable from text alone by evaluating every model in blind mode (no image; see Sec. 5). We report blind accuracy per task as a first-class diagnostic and treat full − blind as the genuinely vision-attributable signal. During this audit we identified one additional attribute task (green-space presence) that was answerable from the option text alone—a model reached 100% accuracy on it with no image, because the correct option wording was systematically identifiable independent of its A/B/C/D position. We *exclude that task entirely* from all results below; the study reports the eight attribute tasks that survive the audit. We note that across the retained tasks the correct-option letter is near-uniformly distributed (A/B/C/D = 743/766/703/720 over the 2932 eight-task benchmark questions), so there is no exploitable positional regularity, and the correct option is not systematically the longest or most detailed (Sec. 6).

Table 1: Benchmark test set (eight attribute tasks, held-out 10% of locations per city). Each question has four options and a single correct letter; image mode is a single marked satellite tile plus four street-level views.

Property	Value
Cities	40
Locations (test)	421
Questions (test)	3135
Attribute tasks	8
Options per question	4
Satellite views / location	1
Street-level views / location	4
Questions per task	
land_use / urban_density / road_type	421 each
amenity_richness / transit_density	421 each
junction_type	334
road_surface	303
building_height	190

Statistics. Table 1 summarizes the benchmark. The eight-task test set comprises 2932 questions over 421 locations across 40 cities; each location supplies one satellite tile and four street-level views.

4 Experimental Setup

Models. Our primary model is **Qwen3.5-9B** fine-tuned on the benchmark's training split with LoRA [4] (rank 16, $\alpha = 16$, applied to all linear layers including the vision tower) and evaluated on the held-out benchmark; we train one model per split (per-city and seen/unseen) and report both. This is the cleanest setting for the view question because the model has been explicitly optimized on the task distribution—so a failure to use both views cannot be attributed to a lack of task alignment. To test that the finding is not specific to our model, we additionally evaluate four *off-the-shelf* remote-sensing VLMs zero-shot on the overhead half of the benchmark: GeoChat-7B [7], SkySenseGPT-7B [15], LHRS-Bot-Nova [9], and EarthDial-4B (RGB) [20]. Table 2 lists their architectures and modalities; all four are single-image, overhead-only models.

Image-ablation protocol. Every question is evaluated under four input conditions: `blind` (prompt only), `sat` (the marked satellite tile only), `sv` (the four street-level views only), and `full` (satellite tile + four street-level views). The text of the question and options is byte-identical across conditions; only the set of supplied images changes. We decode greedily and parse a single answer letter. For the trained model we use the same image preprocessing as training (max edge 512 px).

Table 2: Off-the-shelf remote-sensing VLMs evaluated zero-shot on the overhead (satellite-only) half of the benchmark. All four are single-image and *overhead-only*; none ingests ground-level imagery. "OSM data" marks whether the model's training corpus is derived from OpenStreetMap (relevant to the contamination caveat in Sec. 6).

Model	Base LLM	Vision encoder	Input res.	OSM data
GeoChat-7B [7]	Vicuna-1.5-7B	CLIP ViT-L/14	504	No
SkySenseGPT-7B [15]	Vicuna-1.5-7B	CLIP ViT-L/14	336	No
LHRS-Bot-Nova [9]	Llama-3-8B	SigLIP-L/14	336	**Yes**
EarthDial-4B-RGB [20]	Phi-3-Mini	InternViT-300M	tiled	**Yes**

Metrics. We report per-task accuracy in each condition and two derived quantities: *fusion synergy* $= \mathsf{full} - \max(\mathsf{sat}, \mathsf{sv})$ and *vision contribution* $= \mathsf{full} - \mathsf{blind}$. Synergy > 0 indicates the second view adds accuracy a single view cannot; synergy ≈ 0 indicates redundancy. [TODO: Add bootstrap 95% CIs and TOST equivalence margin once the per-item re-analysis is run (Sec. 5).]

Unified harness. All models—trained and zero-shot—are routed through one evaluation harness with a shared multiple-choice prompt, shared image rendering, and shared answer-letter parsing, so the question text and images are identical across models and only the forward pass differs. Runs use a single RTX PRO 6000 (96 GB) GPU.

5 Results

5.1 The second view adds little over the better single view

Table 3 reports the trained Qwen3.5-9B on the held-out benchmark, per task, under all four conditions, for both splits. Two patterns stand out. First, full tracks the *better single view* closely: per-task fusion synergy ranges from -1.0 to $+4.5$ points and averages $+1.2$/$+1.3$ points (per-city / seen-unseen) across the eight tasks. The gain is positive but small, and street-level imagery alone (sv) already matches or exceeds the two-view full condition on several tasks (land_use, road_type, road_surface). Whether this two-to-three-point margin reflects genuine fusion or merely the model defaulting to the stronger view is the question we resolve at the item level in Sec. 5.5. Second, vision nonetheless matters: $\mathsf{full} - \mathsf{blind}$ is large and positive (e.g. $+22$ on transit_density, $+32$ on amenity_richness), so the model reads the imagery—it simply extracts a single view's worth of information from it.

5.2 The same models fuse when fusion is required (positive control)

A reader should ask whether our ablation could detect fusion at all—perhaps the small attribute-task synergy reflects a measurement that is simply insensitive to

Table 3: Trained Qwen3.5-9B on the held-out benchmark, per attribute task and input condition (accuracy, %). **Syn.** = full − max(sat, sv) is fusion synergy. Synergy is small on every task under both splits; full stays close to the better single view.

	per-city split					**seen/unseen split**				
Task	blind	sat	sv	full	Syn.	blind	sat	sv	full	Syn.
land_use	68.6	75.5	75.1	75.3	−0.2	63.4	72.0	72.4	73.9	+1.4
building_height	45.3	67.4	66.3	66.8	−0.5	50.5	63.7	66.3	66.8	+0.5
urban_density	54.6	61.8	62.7	65.1	+2.4	53.2	71.5	64.8	74.6	+3.1
junction_type	61.4	73.4	68.6	75.1	+1.8	62.0	74.0	71.3	76.3	+2.4
amenity_richness	26.4	53.4	56.5	58.7	+2.1	25.7	53.0	55.6	55.6	+0.0
road_type	68.6	72.4	77.4	76.5	−1.0	68.6	69.6	76.7	76.7	+0.0
road_surface	95.7	95.0	93.4	95.7	+0.7	95.7	95.0	95.4	95.0	−0.3
transit_density	28.3	43.7	45.6	50.1	+4.5	26.4	45.8	44.9	49.4	+3.6
Mean	56.1	67.8	68.2	70.4	+**1.2**	55.7	68.1	68.4	71.1	+**1.3**

Table 4: Positive control: the same trained Qwen3.5-9B (per-city) on cross-view tasks that require both views by construction. Synergy is large and positive, confirming the ablation detects fusion when it is needed.

Cross-view task	blind	sat	sv	full	Syn.
camera_direction	25.2	24.9	–	42.3	+17.3
mismatch_binary_easy	46.6	51.1	52.7	97.4	+44.7
mismatch_binary_hard	55.8	45.8	44.4	90.7	+44.9
mismatch_mcq_easy	20.7	26.4	23.5	96.2	+69.8
mismatch_mcq_hard	22.3	23.3	22.8	86.0	+62.7

the second view. It does not. Table 4 reports the identical model and harness on the cross-view control tasks, whose answers require comparing the two views. Here synergy is large—+17 to +70 points—because a single view is, by construction, insufficient. The contrast with Tab. 3 is the crux of the paper: the same instrument that registers a two-to-three-point gain on the attribute tasks registers a forty-to-seventy-point gain the moment the task genuinely requires both views. The marginal synergy on attributes is therefore a property of those tasks, not an artifact of an insensitive test.

That the cross-view tasks genuinely encode cross-view information—rather than being solved by some shortcut our VLM happens to exploit—is corroborated by an independent, purpose-built model. We evaluate Sample4Geo [1], a Siamese ConvNeXt trained for street↔satellite retrieval, on the mismatch family by embedding the satellite tile and each street-view set and scoring by cosine similarity. This is a different model class and metric from our MCQ VQA and the numbers are not comparable to Tab. 4; we report it only to confirm the tasks' nature. Transferred zero-shot (its VIGOR checkpoint, the closest urban domain), it reaches 0.65 accuracy on the binary same-place/different-place questions—well

Table 5: Off-the-shelf RS-VLMs, zero-shot, single marked satellite tile, per-task accuracy (%) over the eight-task benchmark ($n = 2932$). Last column: our trained Qwen3.5-9B in satellite-only mode (per-city), the like-for-like single-overhead-view reference. LHRS-Bot-Nova and EarthDial are trained on OSM-derived text and their scores should be read as an *upper bound* on visual skill (see Sec. 6).

Task	GeoChat	SkySenseGPT	LHRS-Nova	EarthDial	*Ours (sat)*
land_use	48.7	53.4	46.3	47.5	*75.5*
building_height	30.5	34.7	28.9	32.6	*67.4*
urban_density	13.5	14.5	17.8	30.4	*61.8*
road_type	10.7	18.1	26.1	46.1	*72.4*
road_surface	75.6	80.2	58.7	58.4	*95.0*
junction_type	8.4	30.8	35.3	52.7	*73.4*
amenity_richness	13.5	12.6	17.8	25.4	*53.4*
transit_density	21.4	26.1	34.2	29.7	*43.7*
Overall	26.2	32.0	32.4	**39.9**	***66.6***

above the 0.50 chance rate—so a model that only measures cross-view similarity already carries real signal on these items. It is near chance on the four-way variants (0.27–0.31 vs. 0.25), reflecting how much harder it is to single out the correct nearby scene from three plausible distractors across the viewpoint gap. The attribute tasks, by contrast, offer no such cross-view signal to recover.

5.3 The finding holds zero-shot across off-the-shelf RS-VLMs

To rule out that the redundancy is an artifact of our particular model, we evaluate four off-the-shelf remote-sensing VLMs zero-shot on the overhead (satellite-only) half of the benchmark (Tab. 5). None was trained on our task, our four-way MCQ format, or our marker overlay. They span CLIP- and SigLIP-based encoders, three LLM families, and single-crop versus tiled preprocessing. The trained Qwen3.5-9B in its satellite-only mode is the like-for-like single-overhead-view comparison; it outperforms the best zero-shot model (EarthDial, 39.9%) by +26.7 points overall, as expected for a model fine-tuned on the distribution. The four zero-shot models are satellite-only and therefore inform the single-view baseline and the contamination discussion (Sec. 6), not the synergy claim itself, which is a within-model comparison.

5.4 No prompting strategy recovers complementary signal

If the two views *were* complementary but the model simply failed to combine them, a stronger reasoning procedure should surface the latent signal. It does not. Table 6 reports three strategies—plain decoding, a 16k-token explicit-thinking budget, and a four-turn *observe→reason→self-check→commit* conversation—on the eight attribute tasks, here using an AWQ-quantized Qwen3.5-9B so the

Table 6: Prompting strategies on the eight attribute tasks (Qwen3.5-9B-AWQ, $n = 2932$, per-task means). **Syn.** = full−max(sat, sv); **Vis.** = full−blind. Reasoning improves accuracy but not synergy—the second view remains redundant, and street-view alone (sv) is the single strongest condition in every strategy.

Strategy	blind	sat	sv	full	Syn.	Vis.
plain	41.0	46.5	54.2	51.7	−2.8	+10.8
think-16k	41.4	48.2	54.0	52.5	−2.0	+11.2
multistep (4-turn)	43.4	50.3	55.0	53.2	−2.1	+9.8

long-context reasoning runs fit a single GPU. More deliberation raises accuracy modestly—the model gets better at each task in every view—but synergy stays *negative* throughout, and indeed sv alone is the single strongest condition under every strategy. The extra reasoning accrues to the single-view conditions just as much as to full, so what improves is the model's reading of one view, not its ability to combine two. This mirrors the item-level finding of Sec. 5.5: the obstacle is adjudicating between redundant views, which more thinking does not remove.

5.5 Item-level analysis: the second view interferes more than it helps

A small positive average could still hide genuine fusion on a subset of items, or could be an artifact of the model defaulting to its stronger view. We resolve this by joining the predictions of all four conditions at the item level (every benchmark question carries a stable identifier shared across conditions) and analyzing the 2932 fully paired eight-task items directly.

A per-item single-view oracle beats the two-view model. For each item we define an *oracle* that is allowed to pick the better of the two single-view answers—a model that always chooses the right view, but never fuses. Table 7 compares the two-view full condition against this oracle. The two-view model trails the per-item single-view oracle by 9.3 points (per-city) and 7.9 points (seen/unseen), with paired bootstrap 95% confidence intervals that exclude zero by a wide margin ($[-10.4, -8.2]$ and $[-9.1, -6.8]$, 10,000 resamples). Moreover, the fraction of items solved by *either* single view (the union ceiling) equals the oracle exactly: there is no pool of items that only the *combination* of views answers correctly. In other words, the two views carry essentially no answer-relevant information beyond their better single view, and a system that simply routed each question to the right view would outperform one that sees both.

The second view overturns more correct answers than it rescues. The mechanism behind the oracle gap is an asymmetry in how the second view

Table 7: Item-level analysis on the 2932 paired eight-task questions. **Oracle** picks the better single-view answer per item; **Union** is the fraction either single view answers correctly. **full−oracle** and **synergy** carry paired bootstrap 95% CIs (10,000 resamples). **Help/Hurt** are the percentages of items on which the second view flips the better single-view answer right and wrong, respectively. The two-view model trails the per-item single-view oracle on both splits.

Split	full	oracle	full−oracle	synergy	Help	Hurt
per-city	69.5	78.8	-9.3 $[-10.4, -8.2]$	$+2.2$ $[0.9, 3.3]$	0.9%	10.2%
seen/unseen	70.3	78.2	-7.9 $[-9.1, -6.8]$	$+2.8$ $[1.5, 3.7]$	1.2%	9.1%

changes answers. Relative to the better single view per item, adding the second view corrects a previously wrong answer on only 0.9% / 1.2% of items (per-city / seen-unseen) but flips a previously correct answer to wrong on 10.2% / 9.1%—roughly a ten-to-one ratio against. The model does not reliably defer to whichever view is right; when the views disagree, the second view drags down items the better view had already answered. This is consistent with the prompting results of Sec. 5.4: no amount of explicit reasoning recovers a complementary signal, because the limiting factor is adjudication between redundant views, not reasoning depth.

Aggregate synergy is small and positive, not zero. For completeness, Tab. 7 also reports the pooled fusion synergy with its bootstrap CI: $+2.2$ $[0.9, 3.3]$ (per-city) and $+2.8$ $[1.5, 3.7]$ (seen/unseen). These pooled values are item-weighted and thus slightly exceed the $+1.2/+1.3$ unweighted task means of Tab. 3, because the larger tasks happen to carry the larger synergies. We do not claim the gain is zero—the interval excludes zero. We claim only that it is small: an order of magnitude below the $+40$ to $+70$ points the same model gains on the cross-view control tasks (Tab. 4), and net-negative against optimal single-view selection. Whether a $\sim$2-point average gain justifies acquiring, storing, and serving a second imagery modality is the practical question we take up in Sec. 7.

6 Robustness and Confounds

[TODO: Each item below pairs an objection with the concrete check that addresses it; checks marked [planned] require limited new inference.]

Satellite resolution. Our claim is bounded to commonly served resolutions, but most cities here are covered by *sub-meter* aerial orthophotos (NAIP $\sim$0.6 m/px, IGN $\sim$0.2 m/px, ESRI $\sim$0.3 m/px to 0.5 m/px); Sentinel-2 ($\sim$10 m/px) is only a fallback. The redundancy is thus measured with genuinely high-resolution overhead imagery for the majority of locations, not a coarse proxy. For tasks that are structurally ground-level (e.g. road surface, signage-driven amenities), no overhead resolution recovers the cue; for tasks where higher GSD might plausibly help (building height, urban density) we report a targeted high-resolution sub-study [planned].

The satellite marker. The satellite tile carries a location marker. We verify the results are not an artifact of it via an unmarked-tile ablation [planned] and via the attention analysis below.

Single-image vs. four-view asymmetry. sv supplies four images and sat one, so part of sv's strength could be image count rather than viewpoint. We test a single street-level angle versus the four-view set [planned]; our thesis is explicitly that the *cheaper-to-serve single view suffices*, so street-level richness is part of why it does, not a flaw in the comparison.

Text-only leakage. We report blind accuracy per task (Sec. 5) as the empirical ceiling of what is guessable without images, which subsumes positional, option length, and prior effects. The one task that was fully text-solvable was removed (Sec. 3); we flag remaining tasks with elevated blind accuracy (notably road surface) and rank tasks by vision contribution rather than raw accuracy.

OSM provenance / contamination. Our labels are OSM-derived. Two of the four zero-shot models—LHRS-Bot-Nova and EarthDial—are trained on OSM-derived text (LHRS-Align captions OSM tags across 9259 cities; EarthDial's pretraining QA is built from OSM-labeled imagery), so their scores may reflect shared OSM priors and we treat them as an upper bound on visual skill. GeoChat and SkySenseGPT are *not* OSM-derived. Critically, the synergy claim is a *within-model difference* (full vs. single view), so any uniform OSM prior cancels in the difference. We also report whether removing geographic identifiers from the prompt changes blind accuracy [planned].

Interpretability (corroboration only). Our view-importance claims rest on *causal* input ablation, not attention. As corroboration, we measure the cross-modal attention mass the model places on the satellite token versus the street-level tokens when committing to an answer, reported per task and per layer with normalization for image-token count (a single satellite image vs. four street-level images), and explicitly framed as consistent-with rather than evidence-of importance, in line with known limits of attention as explanation [6, 23]. [TODO: Insert the normalized attention-vs-causal-contribution correlation once computed from the existing attention logs.]

7 Discussion

The picture that emerges is not that overhead imagery is useless, nor that fusion never helps. It is that complementarity is a property of the *task*. When a question is answerable from a single location's imagery—what land use is here, how tall are the buildings, is the road paved—the two views encode largely the same answer-relevant information, and a model reads it off whichever view is more convenient. When a question is *about the relationship* between the views—does this photo belong to this tile, which way is the camera facing—the second view is indispensable, and our models exploit it readily. The eight attribute tasks fall in the first regime; the cross-view tasks in the second.

This has a concrete, practical consequence. Street-level and high-resolution satellite imagery differ sharply in acquisition cost, coverage, licensing, and latency. For the attribute families studied here, the second view buys a gain of only about two to three points on average—and, supplied naively, it degrades roughly one item in ten that the better single view already answered. A deployment that serves a single, well-chosen view therefore forgoes very little accuracy while halving its imagery footprint; if both views are available, the evidence favors *routing* each question to one view over feeding the model both. The synergy, oracle-gap, and vision-contribution measurements we use form a reusable instrument for making that call on a new task before committing to dual-view collection.

8 Conclusion

We measured, rather than assumed, the value of a second perspective for urban attribute recognition. Through a controlled four-way image ablation on a multi-city satellite+street VQA benchmark, we found that across model scale, training regime, city split, and prompting strategy the second view adds only a few points over the better single view—and that a two-view model trails a per-item single-view oracle by 8–9 points, overturning correct answers about ten times as often as it rescues wrong ones. The same models gain 40–70 points on cross-view tasks that genuinely require both views, so the effect is a property of the tasks, not of the instrument. Multi-perspective necessity is task-dependent; for these urban-attribute tasks, a single well-chosen view might be enough.

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